

The top corners of the slide feature decorative circular UI elements. On the left, a circular dial with several rectangular segments is visible. On the right, a similar circular dial with more numerous, thinner segments is shown. Both are rendered in a glowing cyan color against the dark background.

ON YOUR OCEAN'S 11 TEAM, I'M THE AI GUY

*Technically Girl

Harriet Farlow



OUR OBJECTIVES



| Hack the Casino

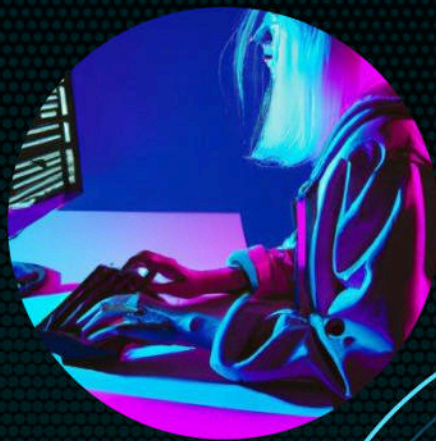
- Casino OSINT and recon
- Hack facial recognition AI
- Reflect on AI Security



WHO AM I



➤ Mileva Security Labs



➤ @HarrietHacks



WHY THE CASINO?



| All AI can be hacked, but people only care when you lose a lot of money..
publically



SOME DISCLAIMERS



- We had Casino Canberra's permission
- These attacks are real but we don't show you how to implement them
- This talk is aimed at cyber folk (don't come at me AI people!)



A BIT ABOUT CASINOS



| For the non-gamblers among us

> They make a lot of money, but have a controversial history



A BIT ABOUT CASINOS

A BIT OF HEAT

Politics

Money launderers are using online gambling sites to disguise shady funds

Fintrac analyzed suspicious transaction reports related to online gambling between 2016 and 2023



Jim Bronskill · The Canadian Press · Posted: Jan 18, 2024 9:52 AM EST | Last Updated: January 18



Australia news

● This article is more than 1 year old

Crown to pay \$450m penalty after failing to comply with anti-money laundering laws

Crown casino and Austrac agree to penalty, which still must be approved by court

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Australian
Associated Press

Tue 30 May 2023 06:05
BST

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A BIT ABOUT CASINOS



| For the non-gamblers among us

- > They make a lot of money, but have been getting a lot of heat
- > They use AI



A BIT ABOUT CASINOS

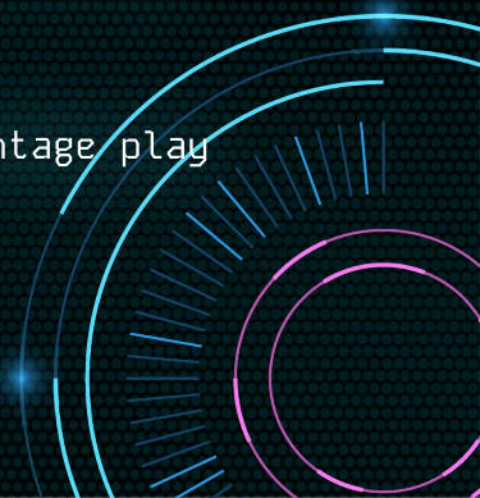


A BIT ABOUT CASINOS



| For the non-gamblers among us

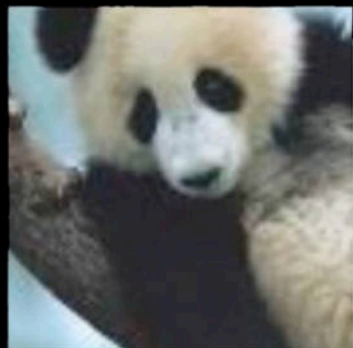
- > They make a lot of money, but have been getting a lot of heat
- > They use AI
- > Card counting is not illegal, but is considered advantage play



HOW TO HACK AN AI



ADVERSARIAL MACHINE LEARNING



"PANDA"

57.7% confidence

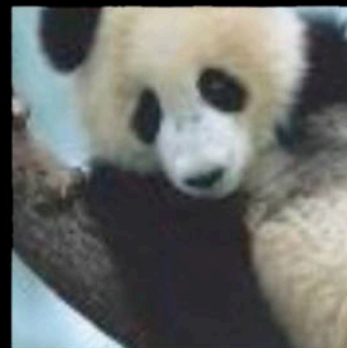
+ .007 x



NOISE

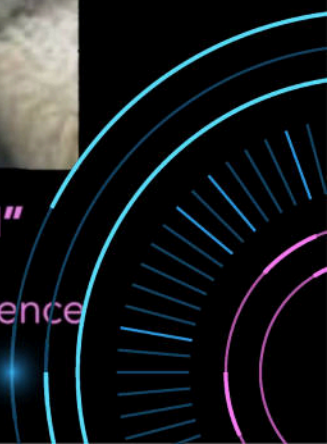
8.2% confidence

=



"GIBBON"

99.3% confidence



ALGORITHM HACKING



RANDOM
NUMBER
GENERATORS

GGB
Global Gaming Business Magazine

playtech CASINO

BIG CIRCLE
MEGA FIRE BLAZE

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Features

SECURITY & SURVEILLANCE: Running the RNG Risk

By [Willy Allison](#) Fri, Jan 20, 2017

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How computers are taking the luck out of gambling



In the late '80s, a real game-changer took place in casinos that would forever change the landscape and opportunities in the gambling industry. Slot machines went from being mechanical devices to being electronic devices.

In order to provide randomness and bigger payouts, a new computerized process was introduced that utilized a random number generator (RNG). The RNG is an algorithm

ALGORITHM HACKING



CARD COUNTING

THE RUNNING COUNT



high cards

Value

-1



low cards

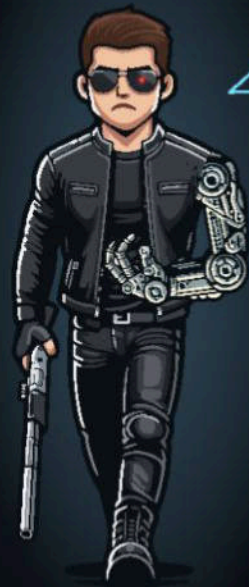
Value

+1



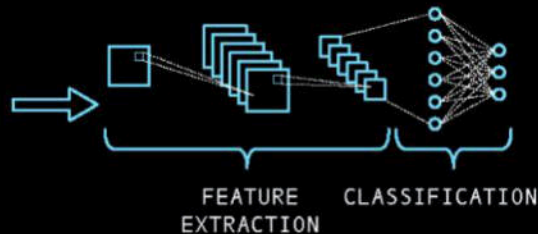
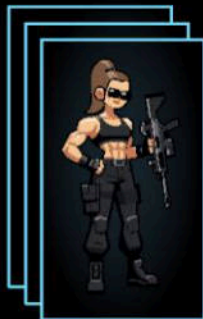
Value

0



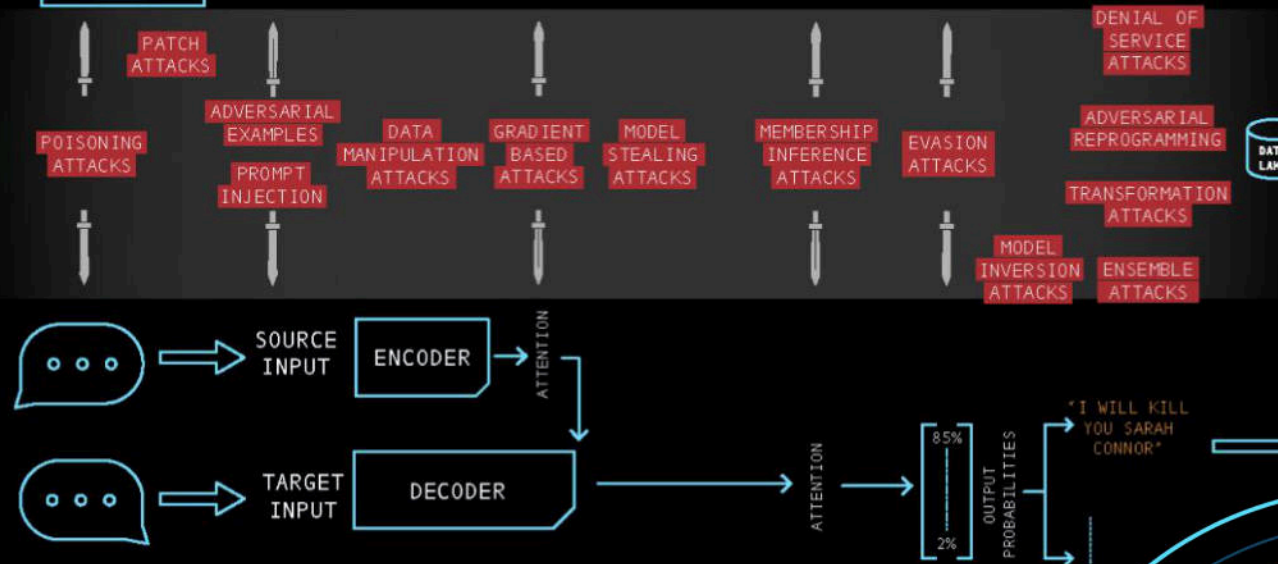
COMPUTER VISION

Convolutional Neural Network



NATURAL LANGUAGE PROCESSING

Transformer



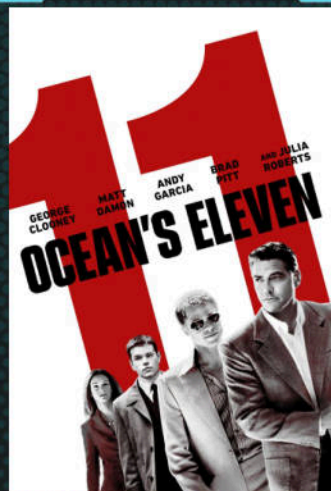
ONTO OTHER
MODELS &
SYSTEMS



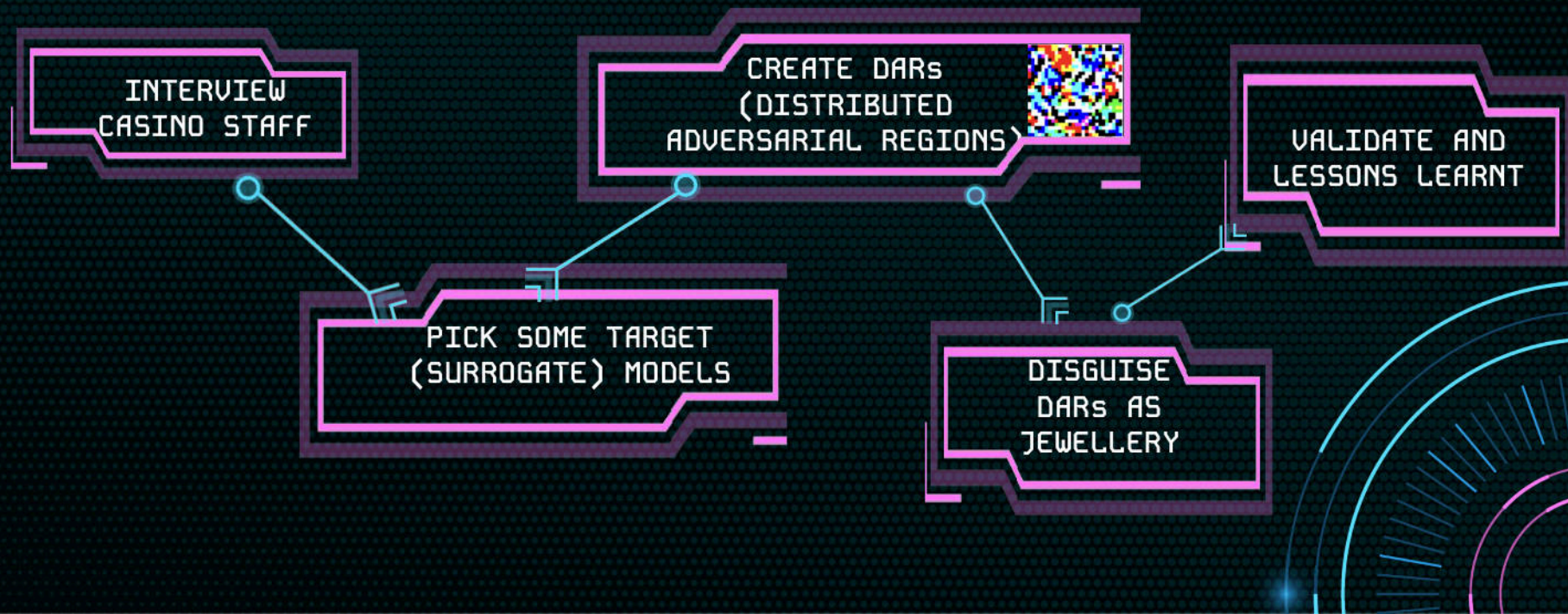
HAS EVERYONE SEEN
OCEAN'S 11?



HEIST 101



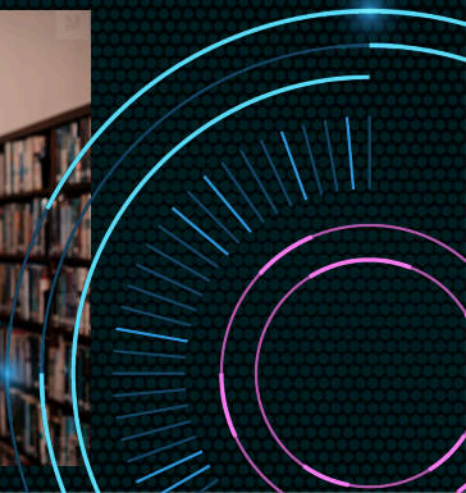
HACKING THE CASINO



STEP 1: RECON



| Tell people I'm a PhD student doing a paper on AI security, and looking to interview experts



STEP 1: RECON



INTERVIEW ANALYSIS



50 interviews

43 organizations

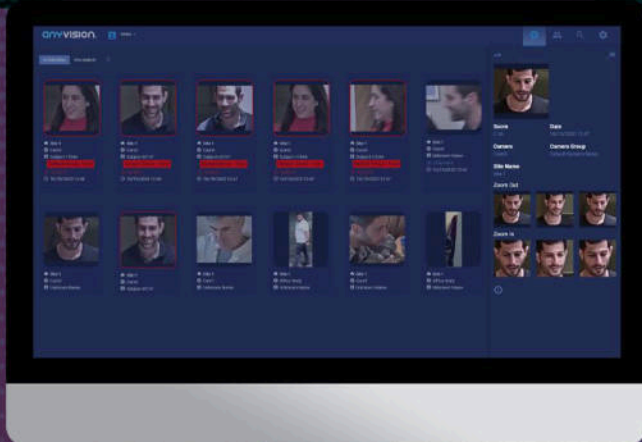
94% could articulate how they **use** AI

8% could articulate how they
secure their AI



CASINOS

FACIAL RECOGNITION



PEOPLE



STEP 2: CHOOSE A VICTIM



VGGFace

DeepFace

DeepID

OpenFace

SFace

FaceNet-123d

FaceNet-512d

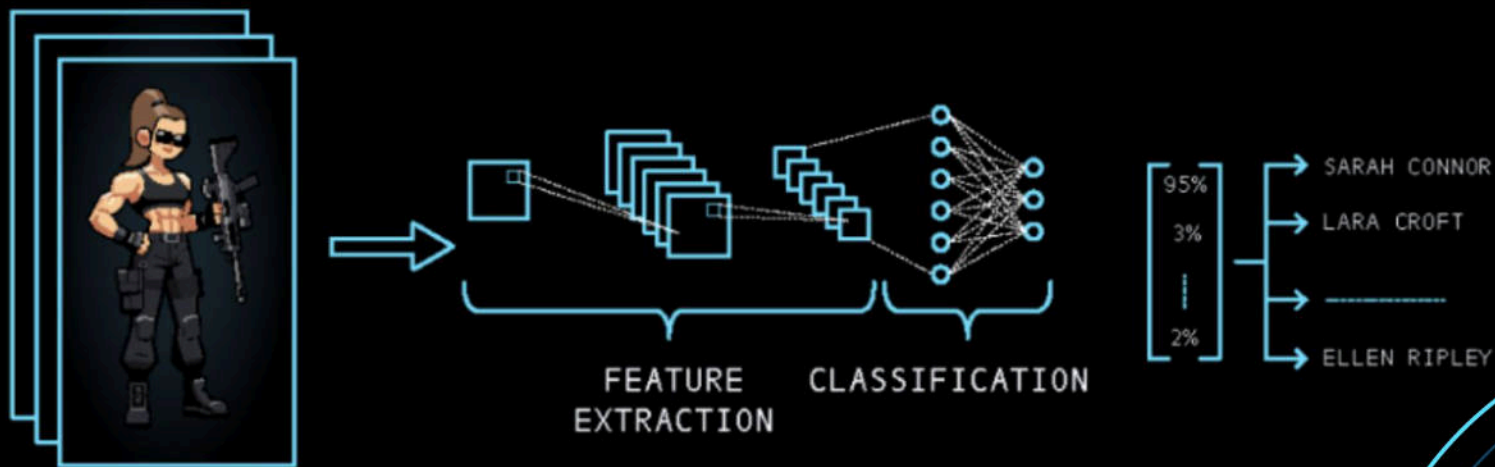
ArcFace

GhostFaceNet

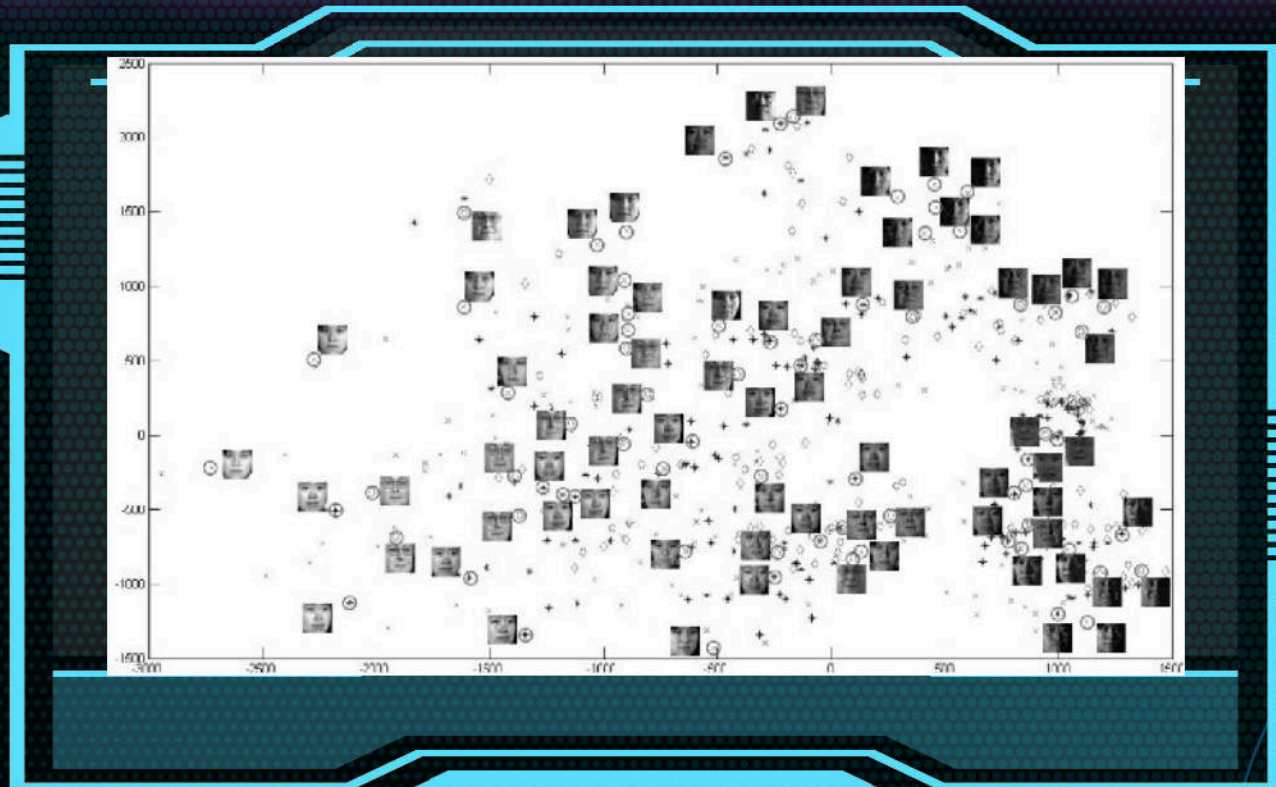
Custom (from scratch)



CONVOLUTIONAL NEURAL NETWORK



EMBEDDING SPACE



MODEL EVALS



DeepFace	FaceNet	VGG-Face	ArcFace	DeepID2+
97.35%	99.63%	98.95%	99.83%	99.47%



95%





CONVERGENCE





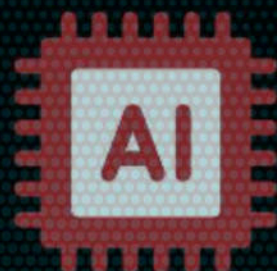
same
data

+



same
training

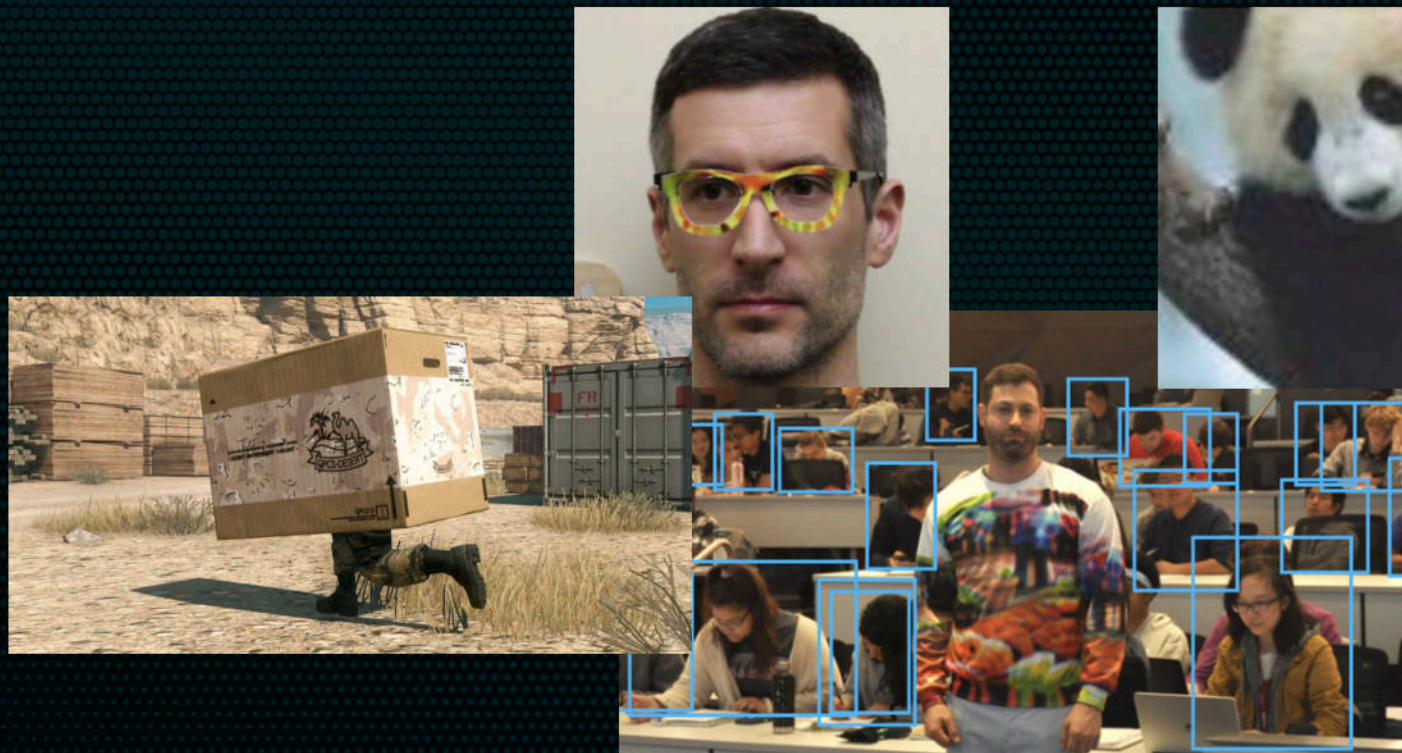
=



same
model



STEP 3: CREATE DARS



STEP 3: CREATE DARS



THE RACE TO ROBUSTNESS: EXPLOITING FRAGILE MODELS FOR URBAN CAMOUFLAGE AND THE IMPERATIVE FOR MACHINE LEARNING SECURITY

Harriet Farlow

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University of New South Wales
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Matthew Garratt

School of Engineering and Information Technology
University of New South Wales
Canberra

Gavin Mount

School of Humanities and Social Sciences
University of New South Wales
Canberra

Tim Lynar

School of Engineering and Information Technology
University of New South Wales
Canberra

26 Jun 2023



1

Identify most influential regions by convolving an n diameter pixel region



Influence measured based on the maximum decrease in confidence based on target model's predictions

2

Apply DARs at regions identified in step 1 and modify DAR parameters

Universal Adversarial Perturbation (UAP)



Projected Gradient Descent (PGD)



Size (n)	2, 6, 8, 10, 14, 18
Number	1, 2, 3, 4

3

Assess the impact of DARs against different benchmarks



	White box	Black box
Models	MobileNet-V2	NasNet-Mobile DenseNet 169
Data	ImageNet	COIL

4

test different shapes, filters and contexts

STEP 3: CREATE DARS



87% YACHT



STEP 3: CREATE DARS



63% BIRD

STEP 3: CREATE DARS



77% CAR

STEP 3: CREATE DARS



87% YACHT



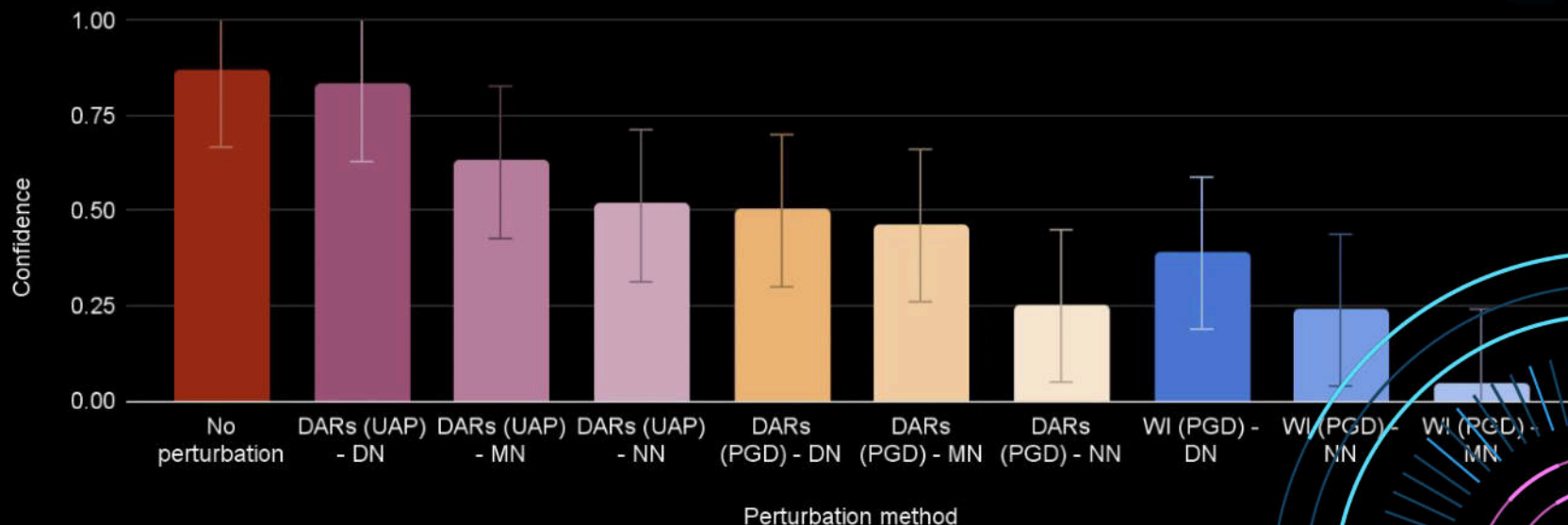
63% BIRD



77% CAR



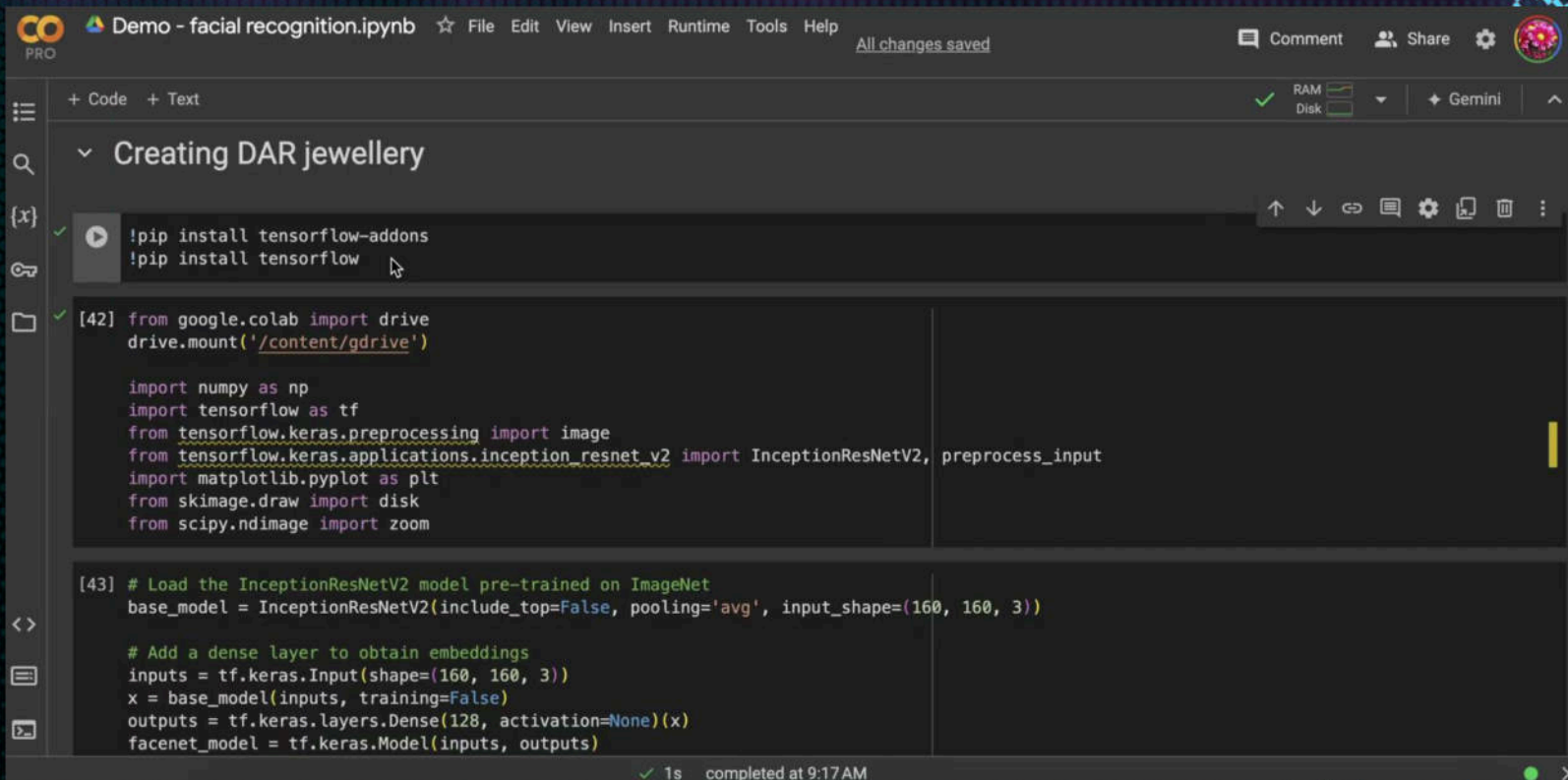
STEP 3: CREATE DARS



STEP 3: CREATE DARS



STEP 3: CREATE DARS



The screenshot shows a Google Colab notebook interface. The title bar reads "Demo - facial recognition.ipynb". The notebook is titled "Creating DAR jewellery". The code is as follows:

```
!pip install tensorflow-addons
!pip install tensorflow

[42] from google.colab import drive
drive.mount('/content/gdrive')

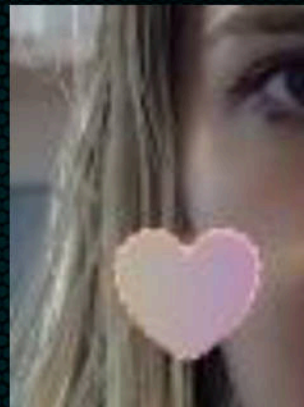
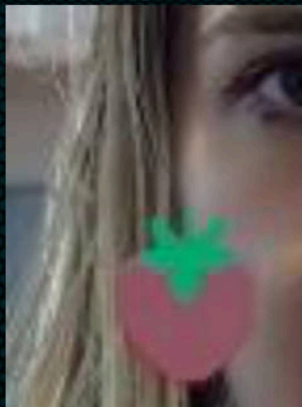
import numpy as np
import tensorflow as tf
from tensorflow.keras.preprocessing import image
from tensorflow.keras.applications.inception_resnet_v2 import InceptionResNetV2, preprocess_input
import matplotlib.pyplot as plt
from skimage.draw import disk
from scipy.ndimage import zoom

[43] # Load the InceptionResNetV2 model pre-trained on ImageNet
base_model = InceptionResNetV2(include_top=False, pooling='avg', input_shape=(160, 160, 3))

# Add a dense layer to obtain embeddings
inputs = tf.keras.Input(shape=(160, 160, 3))
x = base_model(inputs, training=False)
outputs = tf.keras.layers.Dense(128, activation=None)(x)
facenet_model = tf.keras.Model(inputs, outputs)
```

The status bar at the bottom indicates "1s completed at 9:17 AM".

STEP 4: DISGUISE AS JEWELLERY



STEP 4: DISGUISE

CO PRO Demo - facial recognition.ipynb ☆ File Edit View Insert Runtime Tools Help

Comment Share Gemini

+ Code + Text

Facial recognition model

ChatGPT

```
from google.colab import drive
drive.mount('/content/gdrive')
```

[20] !pip install deepface

[21] from deepface import DeepFace
import cv2
from google.colab.patches import cv2_imshow
import os
import matplotlib.pyplot as plt

Paths to your images
known_image_path = '/content/gdrive/MyDrive/Presentations/DEF CON 32 main/Data/Harriet-face/frame_0503.jpg'
unknown_image_path = '/content/gdrive/MyDrive/Presentations/DEF CON 32 main/Data/Harriet-face/Harriet.JPG'
db_path = '/content/face_db/'

Create a database directory if it doesn't exist
os.makedirs(db_path, exist_ok=True)
os.makedirs(os.path.join(db_path, 'Harriet'), exist_ok=True)

Copy the known image to the database directory
known_image_copy_path = os.path.join(db_path, 'Harriet', 'Harriet.JPG')

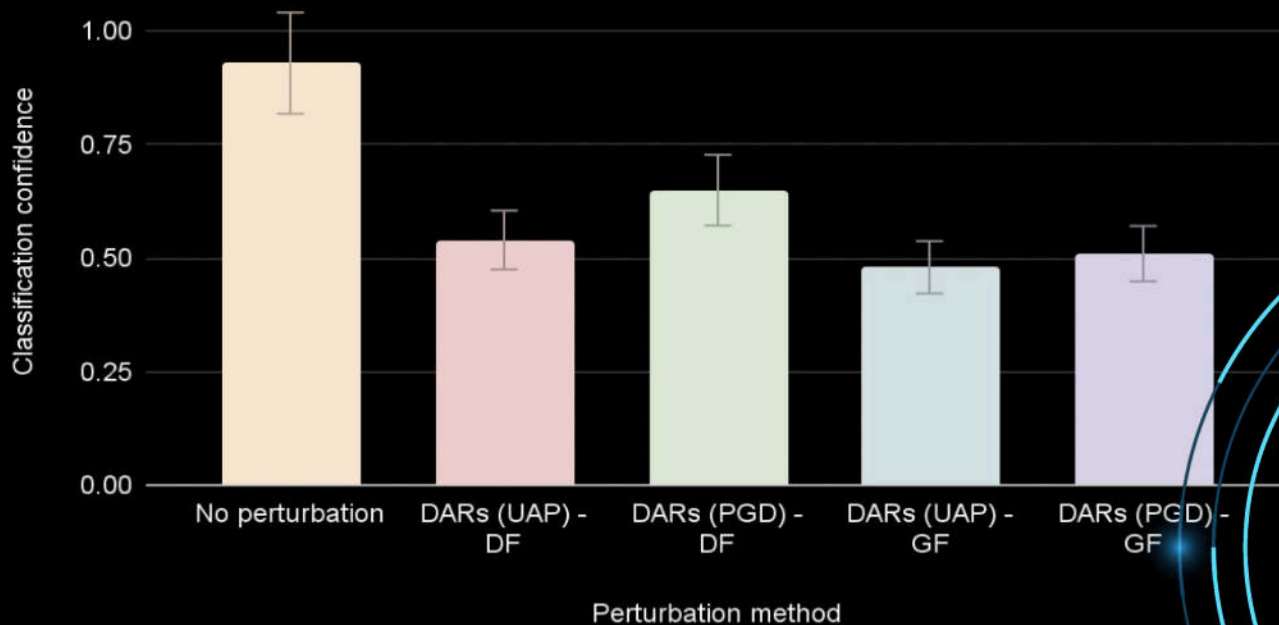
31s completed at 9:05 AM

| Winner winner, chicken dinner



e size
target i

STEP 5: TEST

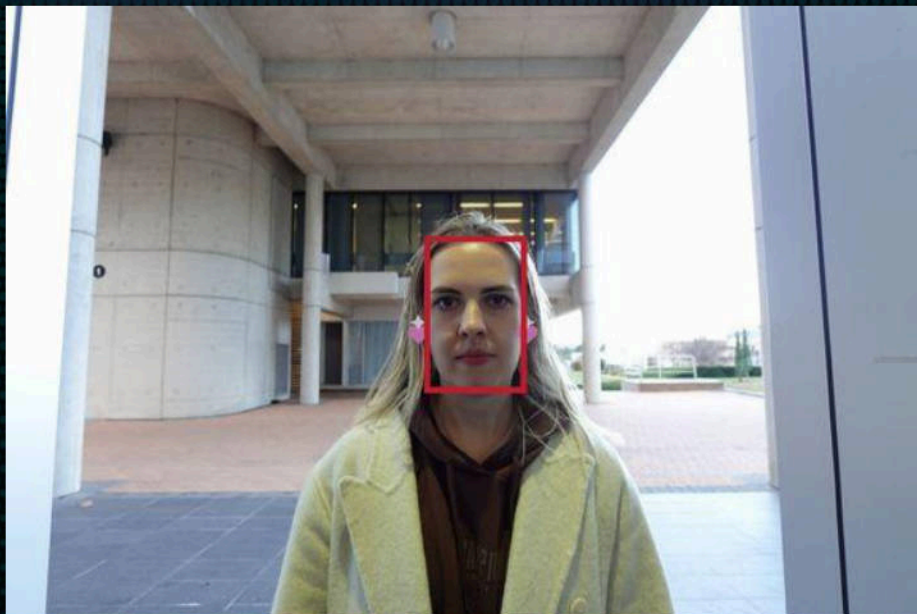




40.4%



STEP 4: DISGUISE AS JEWELLERY



CONTEXT



STEP 5: TEST



STEP 6: BOOK A TRIP





WHAT AM I TRYING TO STEAL?

(What is the target?)





WHAT DOES IT MEAN TO 'HACK' AN AI?

(When the system is inherently
probabilistic?)



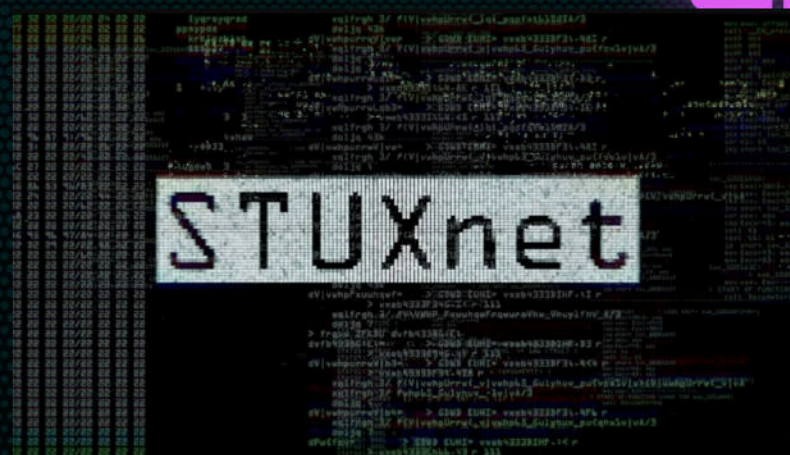


WHAT ARE WE TRYING TO PROTECT?

(What risk management practices
are in place?)



PART OF THE KILL CHAIN





40.4%

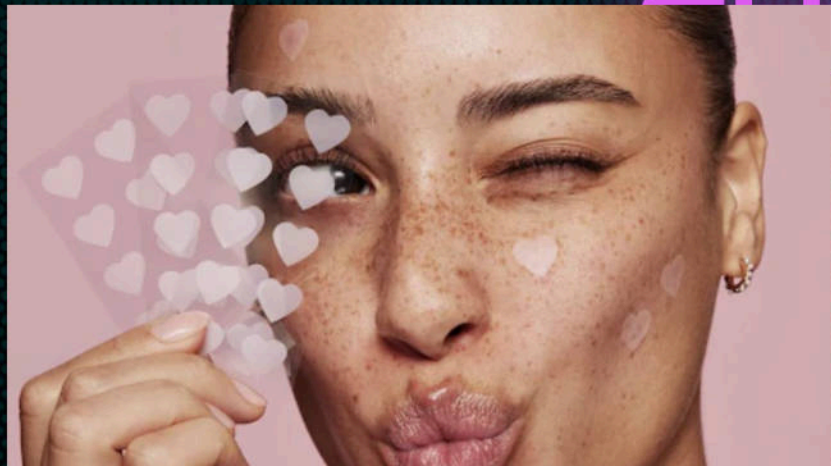




100%

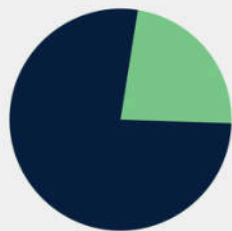


PART OF THE KILL CHAIN





AI SECURITY IS REAL



77%

of companies reported identifying breaches to their AI in the past year. The remaining were uncertain whether their AI models had seen an attack.

30% of IT leaders have deployed a manual defense for adversarial attacks on their existing AI, while just

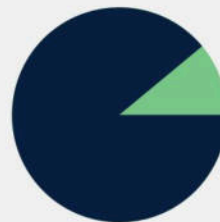
14% are planning and testing for such attacks.

30% Only 30% have deployed technology for model theftjacking, with

20% planning and testing for this threat.



HIDDENLAYER



96%

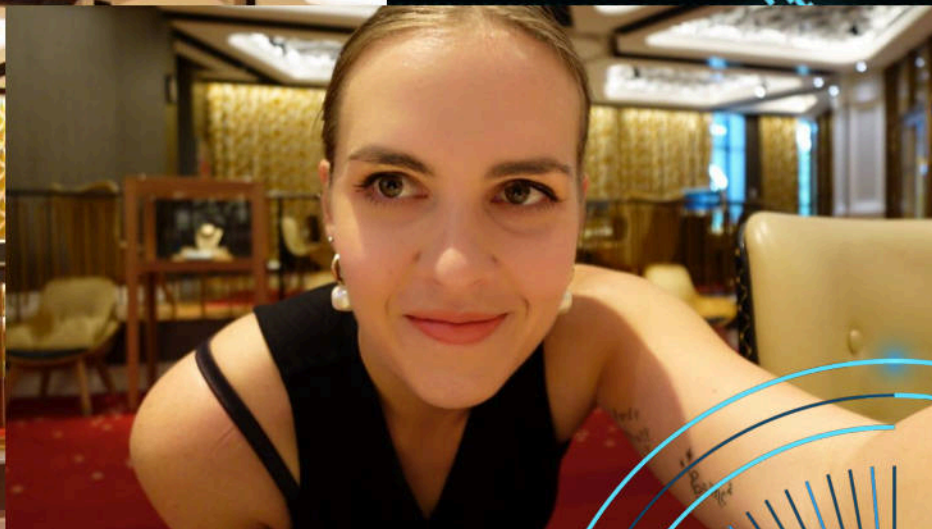
of IT leaders expressed that their AI projects are critical or important to revenue generation over the next 18 months.

~~STEP 7: EVADE FACIAL
RECOGNITION AT~~



STEP 7: PEOPLE AND
PROCESS





SEE MY OTHER TALK



STAY IN TOUCH



@HarrietHacks

I'm also looking for survey participants..

